

Sampark Bhol

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Vellore, India

EXPERIENCE

Software Engineer

Helmit

August 2026 – Present
Munich, Germany / Remote, India

- Build and operate Go, TypeScript and Python microservices on GCP and AWS behind REST and gRPC APIs, owning API contracts, validation, idempotent asynchronous workflows, retries and graceful degradation across distributed systems.
- Scaled a GKE Autopilot service from 3 to 8 replicas with horizontal pod autoscaling and a leader election sharding pattern, and shipped 9 OAuth2 and MTPROTO integrations including an undocumented API reverse engineered from its rejection responses.
- Own CI/CD on GitHub Actions and Cloud Build with build, test and dependency gates, least privilege IAM and WAF policies over 58 PostgreSQL migrations, Redis queues and a Vertex AI agent traced end to end with OpenTelemetry.

Founding Machine Learning Engineer 1

CLR3 Ventures (Spizen Technologies)

Jan. 2026 – August 2026
Bengaluru, India

- Built profitability prediction with ensemble gradient boosting over Apache Kafka streaming pipelines at 92% revenue forecast accuracy, plus a ClickHouse ANN similarity engine and neural recommender serving 10K+ QPS.
- Architected a multi stage LLM agent pipeline for content and synthetic data with retrieval based deduplication and policy filtering, and founded the ML platform on Kafka with Grafana drift and latency monitoring, CI/CD and A/B model rollouts.

Quantum Computing and Research Intern

QuantumX (Appetite Studio)

November 2025 – January 2026
Remote

- Researched hybrid quantum classical models pairing variational quantum circuits with neural networks for combinatorial optimization, designing QAOA and VQE protocols and evaluating post quantum cryptography, and authored Quark, an open source library learning semantic embeddings over QASM circuits on Neon PostgreSQL and RabbitMQ.

AI and Machine Learning Engineer Intern

CSM Technologies

May 2025 – July 2025
Odisha, India

- Shipped production RAG pipelines with LangChain, LangFlow, ChromaDB and Pinecone, lifting domain accuracy 25% through chunking, prompt and retrieval tuning, and built BERT and GPT services for classification, NER and summarization.
- Fine tuned LLMs with LoRA and QLoRA, cutting training compute 60% at baseline parity, then containerized inference with Docker and deployed multi service stacks on Kubernetes.

Deep Learning and Software Development Intern

National Informatics Centre, MeitY, Government of India

May 2024 – July 2024
Odisha, India

- Built anomaly detection with Isolation Forest, DBSCAN, Z-score statistics and graph analysis over 1M+ record government datasets at 95% precision and recall, with predictive models spanning Logistic Regression, SVM, K-Means and PCA.
- Automated Python and SQL ETL and warehousing with schema validation and quality profiling, cutting processing time 70%, and shipped Matplotlib, Seaborn and Plotly dashboards adopted across multiple government departments.

PROJECTS

Cairn (2026): Autonomous experiment platform where coding agents propose, run and score their own repository changes: msgpack RPC daemon, SQLite WAL memory, git worktree isolation, budget caps, anti-Goodhart holdouts, 67+ CI tests.

Browser-LLM (2025): Full transformer inference stack running in the browser on WebGPU with zero backend and no data off device, lazily streaming quantized GGUF and ONNX weights with KV cache management for sub second tokens.

TECHNICAL SKILLS

Languages and Systems: Python, Go, Java, JavaScript, TypeScript, SQL, R; Microservices, REST APIs, gRPC, System Design

AI/ML and Generative AI: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, LangChain, LangGraph, RAG, BERT, GPT, LoRA, QLoRA, PEFT, Embeddings, Semantic Search, NER, OpenCV

Data Science, Analytics and Engineering: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, Tableau, A/B Testing, Hypothesis Testing, Bayesian Inference, Feature Engineering, EDA, Time Series, Kafka, Spark, Airflow, ETL/ELT, Data Warehousing

Cloud, MLOps and Databases: AWS (S3, Lambda, RDS, SageMaker, Bedrock), GCP (GKE, Cloud Run, Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, MLflow, Kubeflow, Grafana, FastAPI, Node.js, React, PostgreSQL, MySQL, MongoDB, Redis, ClickHouse, Elasticsearch, Pinecone, ChromaDB, FAISS, Qiskit, QAOA, VQE, Agile, Git

ACHIEVEMENTS & CERTIFICATIONS

Research: Authored, “Energy Efficient Hybrid WSN Protocol”, published in Elsevier Procedia Computer Science (2025) for novel algorithm development on sensor’s energy efficiency; quantum enhanced ML paper under review.

Open Source: Quark (QuantumX); AI/ML Lead, IEEE Computer Society VIT, 200+ participant hackathons.

Certifications: Oracle Cloud Infrastructure, Oracle GenAI, IBM Deep Learning, NVIDIA Transformer NLP, Saviynt Security